

Regression Diagnostics

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```
library(dplyr)
library(ggplot2)
library(scales)
library(knitr)
library(broom)
library(gridExtra)

load("../data/model_results.RData")
```

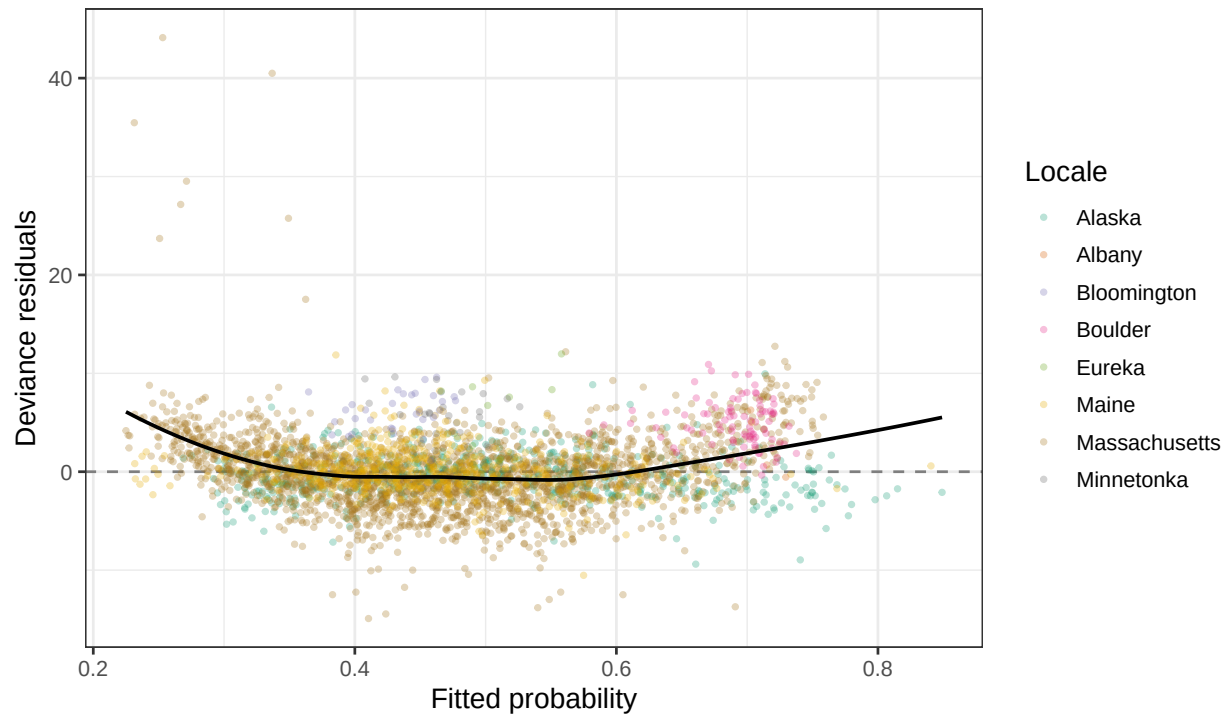
Diagnostics on single-imputation fits (imputation 1) of the quasibinomial GLM with demographic controls (M2). Pooled estimates across all 5 imputations are substantively identical.

1. Deviance residuals vs. fitted

```
diag_data$.dev_resid <- residuals(diag_m2, type = "deviance")
diag_data$.fitted_p <- fitted(diag_m2)

ggplot(diag_data, aes(x = .fitted_p, y = .dev_resid)) +
  geom_hline(yintercept = 0, color = "grey50", linetype = "dashed") +
  geom_point(aes(color = locale), alpha = 0.3, size = 0.8) +
  geom_smooth(method = "loess", se = FALSE, color = "black", linewidth = 0.8) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Fitted probability", y = "Deviance residuals",
       title = "M2: Deviance residuals vs. fitted", color = "Locale") +
  theme_bw(base_size = 13)
```

M2: Deviance residuals vs. fitted

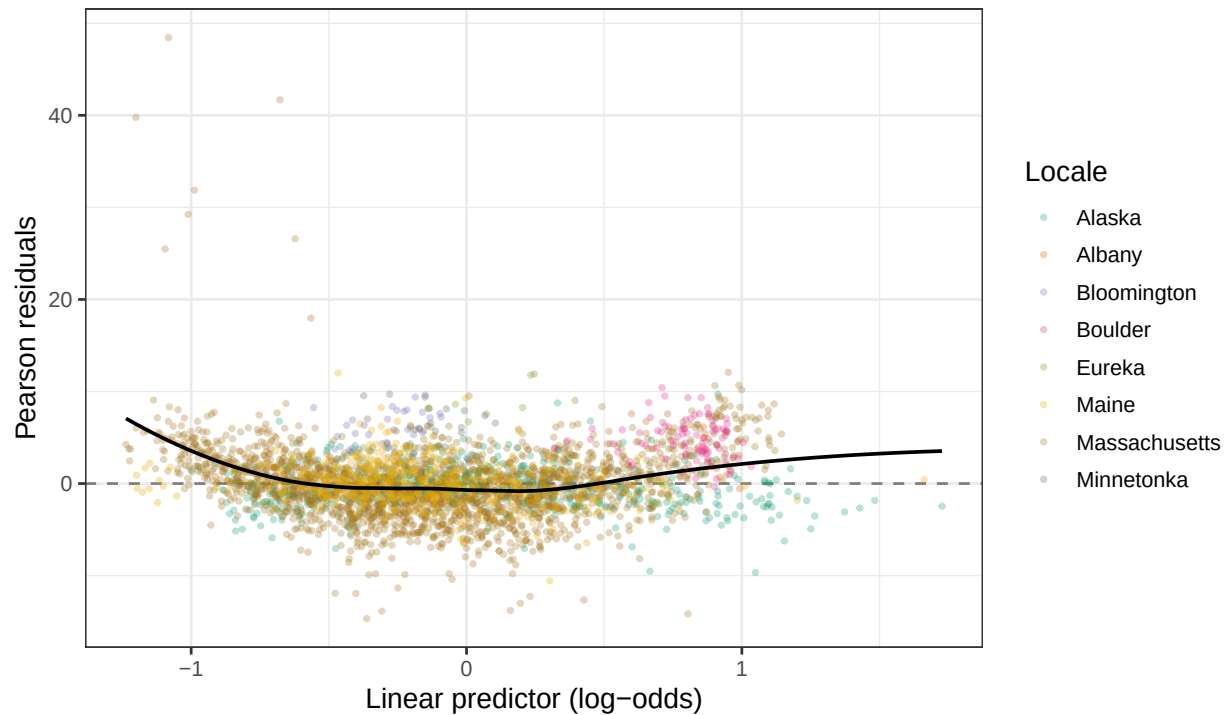


2. Pearson residuals vs. linear predictor

```
diag_data$.pearson <- residuals(diag_m2, type = "pearson")
diag_data$.eta <- predict(diag_m2, type = "link")

ggplot(diag_data, aes(x = .eta, y = .pearson)) +
  geom_hline(yintercept = 0, color = "grey50", linetype = "dashed") +
  geom_point(aes(color = locale), alpha = 0.3, size = 0.8) +
  geom_smooth(method = "loess", se = FALSE, color = "black", linewidth = 0.8) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Linear predictor (log-odds)", y = "Pearson residuals",
       title = "M2: Pearson residuals vs. linear predictor",
       color = "Locale") +
  theme_bw(base_size = 13)
```

M2: Pearson residuals vs. linear predictor



3. Overdispersion

```
disp <- deviance(diag_m2) / df.residual(diag_m2)
cat("Dispersion parameter (deviance/df):", round(disp, 2), "\n\n")
```

Dispersion parameter (deviance/df): 12.75

```
cat("Values >> 1 indicate overdispersion, expected in aggregate vote data\n")
```

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```
cat("(voters within a precinct are correlated, not independent Bernoulli trials).\n")
```

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```
cat("The quasibinomial model corrects standard errors for this.\n")
```

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4. Heteroskedasticity

Variance of Pearson residuals across bins of the fitted value. For a well-specified quasibinomial, this should be roughly constant (around the dispersion parameter).

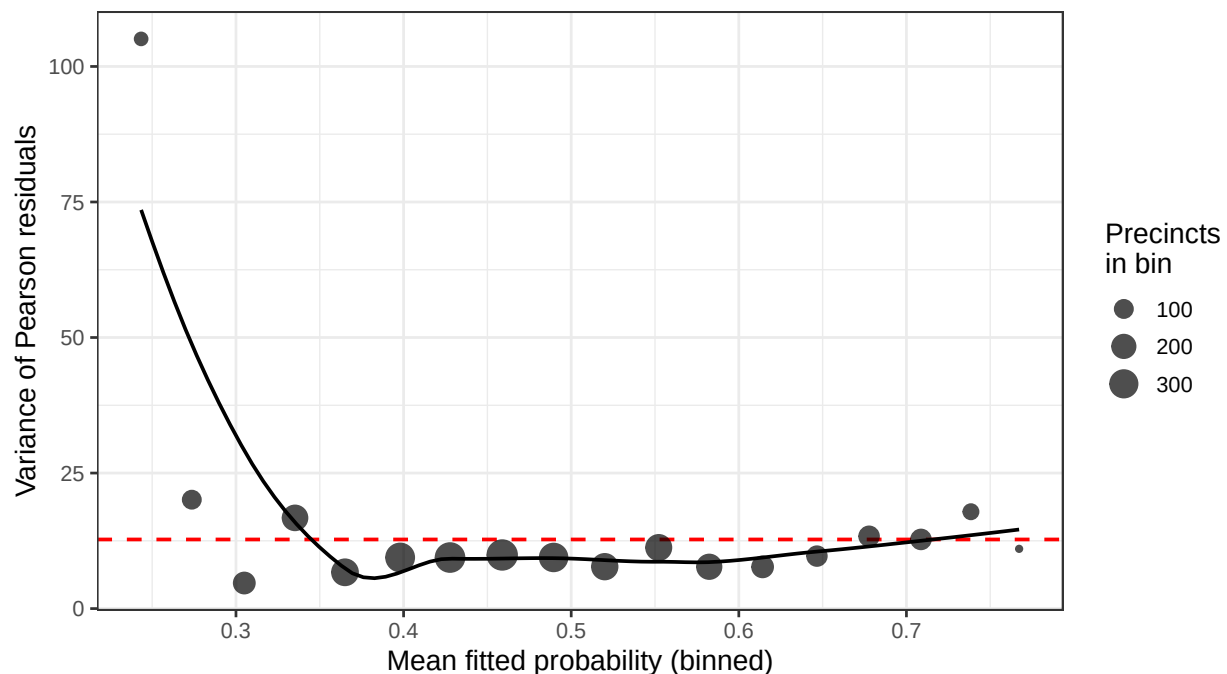
```
diag_data$.fitted_bin <- cut(diag_data$.fitted_p, breaks = 20)

bin_var <- diag_data %>%
  group_by(.fitted_bin) %>%
  summarise(
    mean_fitted = mean(.fitted_p),
    var_pearson = var(.pearson),
    n = n(),
    .groups = "drop"
  ) %>%
  filter(n >= 10)

ggplot(bin_var, aes(x = mean_fitted, y = var_pearson)) +
  geom_hline(yintercept = disp, linetype = "dashed", color = "red",
    linewidth = 0.8) +
  geom_point(aes(size = n), alpha = 0.7) +
  geom_smooth(method = "loess", se = FALSE, color = "black", linewidth = 0.8) +
  labs(x = "Mean fitted probability (binned)",
    y = "Variance of Pearson residuals",
    title = "Heteroskedasticity diagnostic",
    subtitle = "Red line = overall dispersion; flat = homoskedastic",
    size = "Precincts\nin bin") +
  theme_bw(base_size = 13)
```

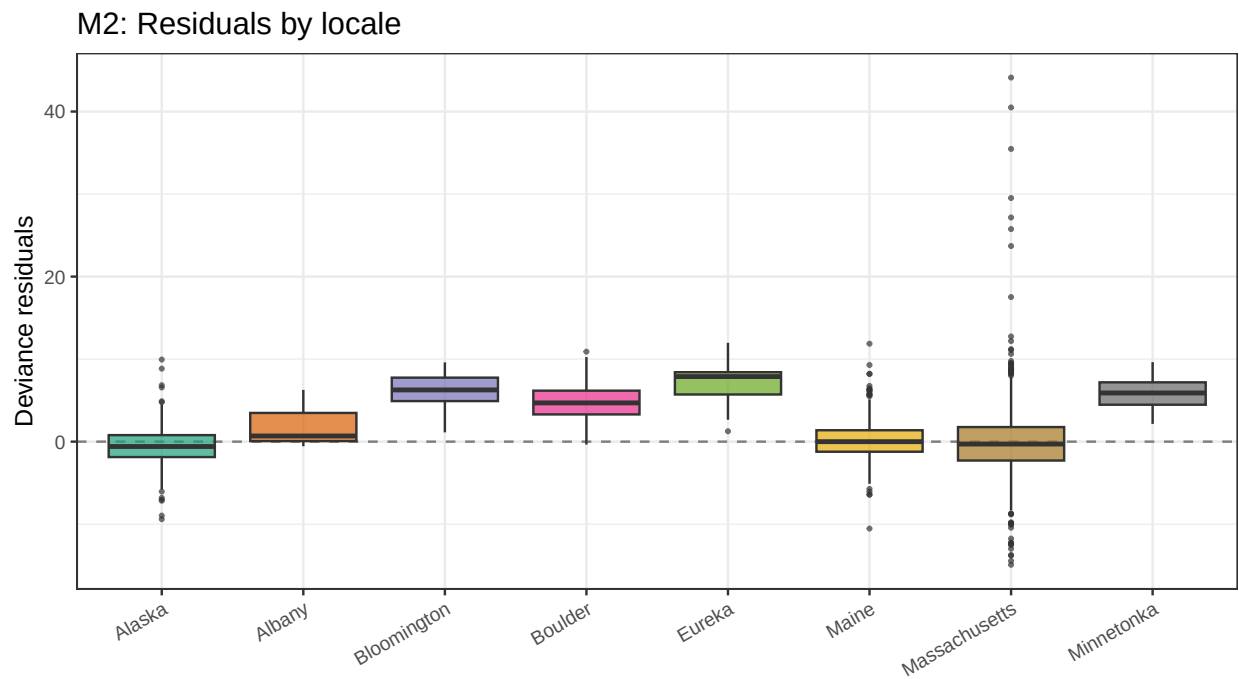
Heteroskedasticity diagnostic

Red line = overall dispersion; flat = homoskedastic



5. Residuals by locale

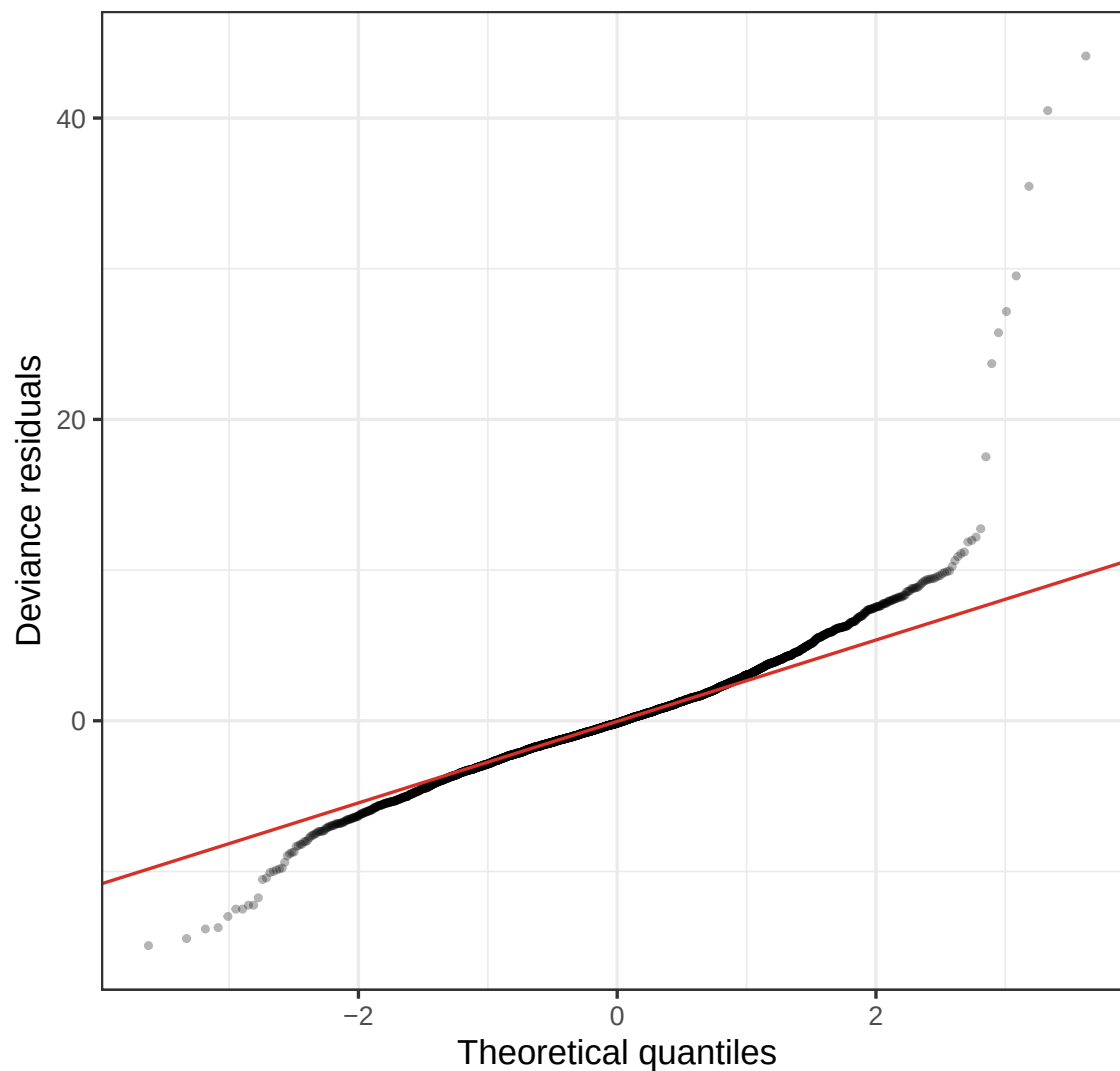
```
ggplot(diag_data, aes(x = locale, y = .dev_resid, fill = locale)) +  
  geom_hline(yintercept = 0, color = "grey50", linetype = "dashed") +  
  geom_boxplot(alpha = 0.7, outlier.size = 0.8) +  
  scale_fill_brewer(palette = "Dark2", guide = "none") +  
  labs(x = NULL, y = "Deviance residuals",  
       title = "M2: Residuals by locale") +  
  theme_bw(base_size = 13) +  
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



6. Q-Q plot

```
ggplot(diag_data, aes(sample = .dev_resid)) +  
  stat_qq(alpha = 0.3, size = 0.8) +  
  stat_qq_line(color = "#D73027") +  
  labs(x = "Theoretical quantiles", y = "Deviance residuals",  
       title = "M2: Normal Q-Q plot") +  
  theme_bw(base_size = 13)
```

M2: Normal Q-Q plot



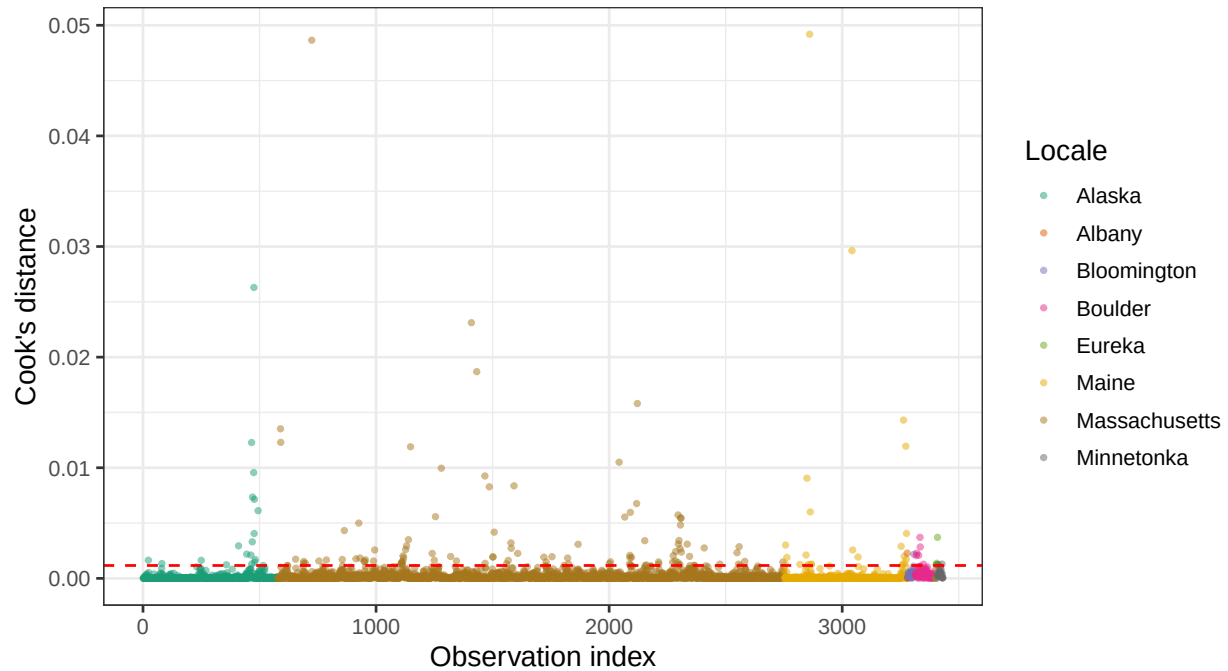
7. Influence: Cook's distance

```
diag_data$.cooks <- cooks.distance(diag_m2)
threshold <- 4 / nrow(diag_data)

ggplot(diag_data, aes(x = seq_along(.cooks), y = .cooks)) +
  geom_point(aes(color = locale), alpha = 0.5, size = 0.8) +
  geom_hline(yintercept = threshold, linetype = "dashed", color = "red") +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Observation index", y = "Cook's distance",
       title = "M2: Cook's distance",
       subtitle = paste("Red line = 4/n =", round(threshold, 5)),
       color = "Locale") +
  theme_bw(base_size = 13)
```

M2: Cook's distance

Red line = $4/n = 0.00117$



```
high_cooks <- diag_data %>%
  filter(.cooks > threshold) %>%
  select(precinct_id, locale, dem_share, yes_share, .fitted_p, .dev_resid, .cooks) %>%
  arrange(desc(.cooks))

cat("Observations above 4/n threshold:", nrow(high_cooks), "/", nrow(diag_data), "\n\n")
```

Observations above 4/n threshold: 154 / 3432

```
high_cooks %>% head(10) %>% kable(digits = 4, caption = "Top 10 influential observations")
```

Table 1: Top 10 influential observations

precinct_id	locale	dem_share	yes_share	.fitted_p	.dev_resid	.cooks
PORTLAND	Maine	0.7397	0.7153	0.7035	4.9972	0.0492
Berkley_-_1	Massachusetts	0.4407	0.7113	0.2531	44.1174	0.0486
BANGOR	Maine	0.5125	0.5327	0.5749	-10.5301	0.0296
District 33 - Absentee	Alaska	0.7809	0.7754	0.7137	9.9482	0.0263
Georgetown_-_1	Massachusetts	0.5248	0.8064	0.3368	40.4946	0.0231
Granville_-_1	Massachusetts	0.4061	0.7690	0.2314	35.4695	0.0187
Provincetown_-_1	Massachusetts	0.9192	0.5588	0.6909	-13.7258	0.0158
LEBANON	Maine	0.2936	0.4906	0.3855	11.8676	0.0143
Acushnet_-_1	Massachusetts	0.4586	0.5801	0.2669	27.1629	0.0135
Acushnet_-_2	Massachusetts	0.4341	0.4953	0.2507	23.7044	0.0123

8. Multicollinearity (VIF)

```
X <- model.matrix(diag_m2)[, -1]
vif_vals <- sapply(seq_len(ncol(X)), function(j) {
  r2 <- summary(lm(X[, j] ~ X[, -j]))$r.squared
  1 / (1 - r2)
})
names(vif_vals) <- colnames(X)

tibble(term = names(vif_vals), VIF = round(vif_vals, 2)) %>%
  arrange(desc(VIF)) %>%
  kable(caption = "Variance Inflation Factors. VIF > 10 = concern.")
```

Table 2: Variance Inflation Factors. VIF > 10 = concern.

term	VIF
recent_lpw	17.95
dem_share:recent_lpw	11.82
dem_share	9.15
pct_bach_plus	6.17
median_income	4.15
pct_white_vap	3.10
pct_renter	2.61
pct_hispanic_vap	1.94
pct_black_vap	1.81

9. Partial-effect plots

Predicted probability of RCV support across the range of each demographic variable, holding other predictors at their medians.

```
key_vars <- c("pct_white_vap", "pct_bach_plus", "median_income", "pct_renter")
var_labels <- c("% White VAP", "% Bachelor's+", "Median Income", "% Renter")

medians <- diag_data %>%
  summarise(across(c(dem_share, recent_lpw, pct_white_vap, pct_black_vap,
    pct_hispanic_vap, median_income, pct_bach_plus, pct_renter),
    median))

plots <- lapply(seq_along(key_vars), function(k) {
  v <- key_vars[k]
  grid <- medians[rep(1, 200), ]
  grid[[v]] <- seq(min(diag_data[[v]], na.rm = TRUE),
    max(diag_data[[v]], na.rm = TRUE),
    length.out = 200)
  grid$pred <- predict(diag_m2, newdata = grid, type = "response")

  ggplot(grid, aes(x = .data[[v]], y = pred)) +
    geom_line(color = "#D73027", linewidth = 1) +
    geom_rug(data = diag_data, aes(x = .data[[v]], y = NULL),
```

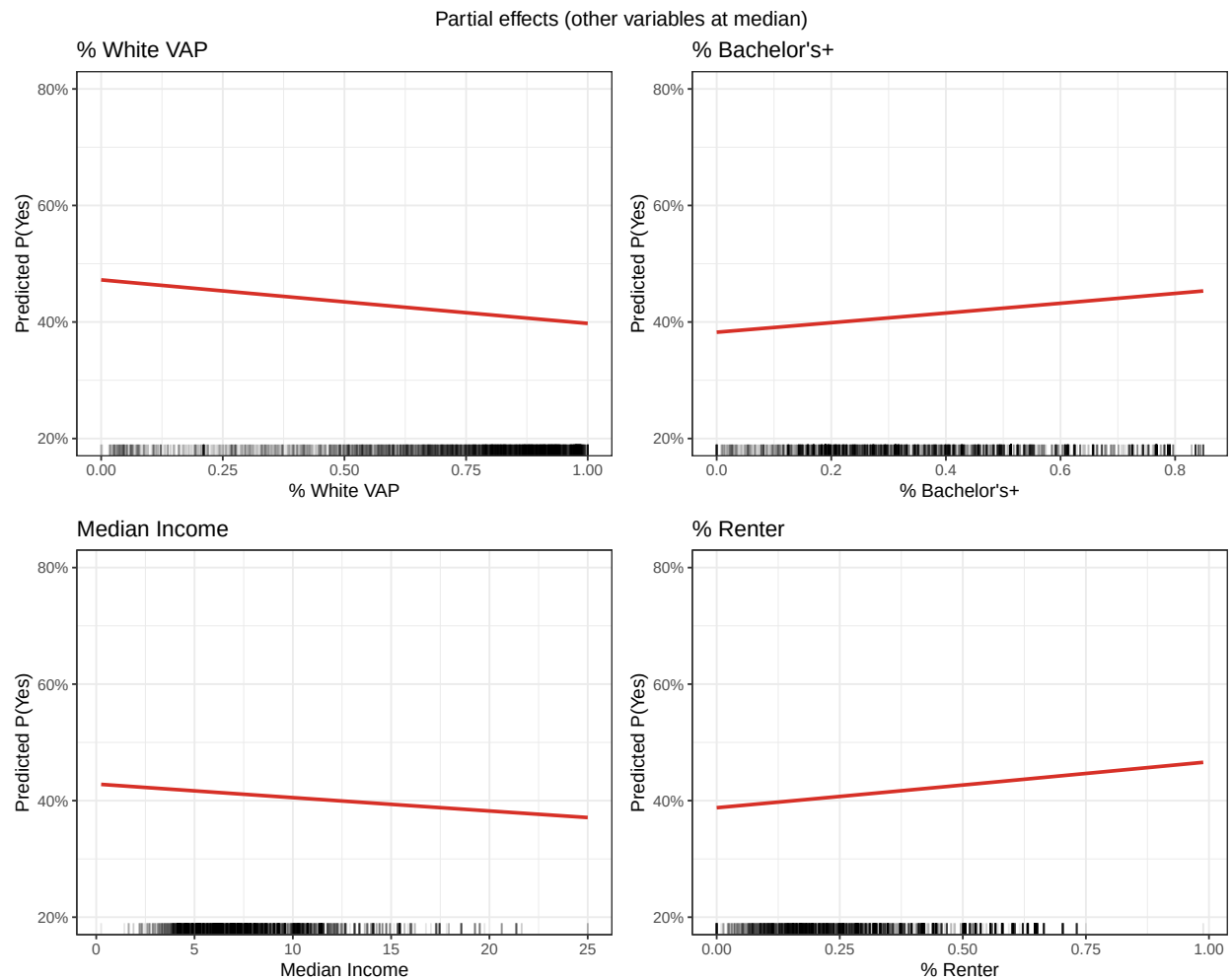


```

    alpha = 0.1, sides = "b") +
    scale_y_continuous(labels = percent_format(1), limits = c(0.2, 0.8)) +
    labs(x = var_labels[k], y = "Predicted P(Yes)",
         title = var_labels[k]) +
    theme_bw(base_size = 11)
  })

gridExtra::grid.arrange(grobs = plots, ncol = 2,
                        top = "Partial effects (other variables at median)")

```



Summary

```

tibble(
  Diagnostic = c(
    "Overdispersion",
    "Influential obs (Cook's D > 4/n)",
    "Max VIF",
    "Heteroskedasticity"
  ),

```

```

Result = c(
  paste(round(disp, 1), "(corrected by quasibinomial)"),
  paste(nrow(high_cooks), "of", nrow(diag_data)),
  round(max(vif_vals), 2),
  "See binned variance plot above"
)
) %>%
kable(caption = "Diagnostic summary")

```

Table 3: Diagnostic summary

Diagnostic	Result
Overdispersion	12.7 (corrected by quasibinomial)
Influential obs (Cook's D > 4/n)	154 of 3432
Max VIF	17.95
Heteroskedasticity	See binned variance plot above